Covariant Emergence of Special-Relativistic Kinematics and Einsteinian Gravitational Dynamics from a Minimal Vector–Tensor Action

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Abstract

We study a minimal covariant vector–tensor framework consisting of the Einstein–Hilbert action, a generally covariant action for a unit timelike vector field u^μ satisfying $u^\mu u_\mu = -1$, and minimally coupled matter. Within this framework we isolate three structural results. First, local special-relativistic kinematics follow from the Lorentzian metric structure in each local inertial frame; no separate postulate of Lorentz invariance is required beyond the assumption of Lorentzian signature. Second, variation of the total action with respect to the metric yields gravitational field equations of Einstein form with covariantly conserved total stress– energy. Third, for minimally coupled structureless test bodies, universality of free fall follows in the test-body limit from matter stress–energy conservation.

The framework is mathematically equivalent to Einstein–Æther and khronometric theories. The individual ingredients of these results are known in that literature; the contribution here is their joint systematic presentation with explicit separation of the structural inputs required for each. In particular, Lorentzian kinematics, Einstein-form dynamics, and weak-equivalence behavior are shown to rest on logically distinct assumptions. We also summarize the post-Newtonian constraint structure inherited from Einstein–Æther theory: $\gamma = \beta = 1$ for the standard asymptotic-alignment boundary condition, while preferred-frame parameters and the gravitational-wave speed furnish the operative observational bounds.

Keywords. Einstein–Æther theory; vector-tensor gravity; relativistic kinematics; weak equivalence principle; parametrized post-Newtonian gravity; preferred-frame effects.

1 Introduction and Scope

Modern relativistic physics rests on three familiar pillars: local special-relativistic kinematics, Einsteinian gravitational dynamics, and the weak equivalence principle (WEP).

Historically these entered through distinct routes and are often treated as logically independent postulates [1, 2, 6]. A natural foundational question is which of these features are structurally enforced by a common covariant action, and which require separate assumptions.

This paper addresses that question within a minimal vector–tensor framework: a Lorentzian metric $g_{\mu\nu}$, the Einstein–Hilbert action, a dynamical unit timelike vector field u^μ, and minimal coupling of matter to the metric alone. This is a standard Einstein–Æther setup [7, 8]. The present contribution is conceptual and organizational: it provides a joint, explicit, and logically decomposed presentation of which assumptions enforce which relativistic properties within one common variational framework.

More specifically, we show that:

(i) local special-relativistic kinematics follow from Lorentzian signature;
(ii) Einstein-form gravitational field equations follow from the Einstein–Hilbert variational structure;
(iii) geodesic free fall for minimally coupled, structureless test bodies follows in the test-body limit from covariant matter conservation.

These three implications depend on distinct structural inputs and should not be conflated.

The paper is thus best read as a foundations-oriented synthesis of known Einstein–Æther results, with an emphasis on logical separation rather than new phenomenology. The phenomenological discussion that is included serves only to locate the framework within the standard post-Newtonian constraint literature [9, 6].

2 Framework and Main Proposition

2.1 Assumptions

Let $(M, g_{\mu\nu})$ be a four-dimensional manifold with Lorentzian metric of signature $(-,+,+,+)$. Consider the action

$$S = S_{\mathrm{g}}[g_{\mu\nu}] + S_u[g_{\mu\nu}, u^\mu] + S_{\mathrm{m}}[g_{\mu\nu}, \psi], \tag{1}$$

where:

- S_{g} is the Einstein–Hilbert action;
- S_u is a generally covariant action for a unit timelike vector field u^μ satisfying $u^\mu u_\mu = -1$;
- S_{m} is the matter action, depending on matter fields ψ and the metric only.

No independent assumption of Lorentz invariance, geodesic motion, or the WEP is imposed beyond Lorentzian geometry, diffeomorphism invariance, and minimal matter coupling.

2.2 Main statement

Proposition 1 (Structural decomposition of relativistic physics)**.** *Under the assumptions above, the following statements hold.*

1. Local special-relativistic kinematics. *At every point $p \in M$ there exist local inertial coordinates in which $g_{\mu\nu}(p) = \eta_{\mu\nu}$ and*

$$d\tau^2 = dt^2 \left(1 - \frac{v^2}{c^2}\right), \tag{2}$$

so that the standard Lorentz factor

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{3}$$

follows directly from the Lorentzian metric structure.

2. Einstein form of the gravitational field equations. *Variation of the total action with respect to $g_{\mu\nu}$ yields*

$$G_{\mu\nu} = 8\pi G \left(T^{(\mathrm{m})}_{\mu\nu} + T^{(u)}_{\mu\nu} \right), \tag{4}$$

with covariant conservation of the total stress–energy tensor.

3. Free fall of minimally coupled structureless test bodies. *If matter is minimally coupled to the metric alone, then in the test-body limit the matter equations imply metric geodesic motion for pressureless, structureless bodies:*

$$U^\mu \nabla_\mu U^\nu = 0. \tag{5}$$

This statement applies to the test-body regime and does not exclude strong-field Nordtvedt-type effects for self-gravitating extended bodies.

Corollary 1.1. *If the vector sector is trivial, e.g. $c_i = 0$ in the standard two-derivative Einstein–Æther action, then $T^{(u)}_{\mu\nu} = 0$ and*

$$G_{\mu\nu} = 8\pi G\, T^{(\mathrm{m})}_{\mu\nu}, \tag{6}$$

so general relativity is recovered as a consistent special case.

Remark 1. *The point of proposition 1 is not that the field content alone guarantees all three properties in any conceivable theory. Rather, within the present minimal variational architecture, these properties arise from distinct and identifiable structural assumptions. It is therefore useful to separate their logical dependence explicitly.*

3 Local Kinematics

At any event $p \in M$, the Lorentzian metric admits local inertial coordinates such that

$$g_{\mu\nu}(p) = \eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1), \qquad \partial_\alpha g_{\mu\nu}(p) = 0. \tag{7}$$

For a timelike worldline,

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu. \tag{8}$$

In local inertial coordinates with $x^0 = ct$, this becomes

$$d\tau^2 = dt^2 - \frac{|d\mathbf{x}|^2}{c^2} = dt^2 \left(1 - \frac{v^2}{c^2} \right), \tag{9}$$

which yields the standard Lorentz factor

$$\gamma = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{10}$$

Defining $U^\mu = dx^\mu/d\tau$ and $p^\mu = mU^\mu$, one obtains

$$g_{\mu\nu}U^\mu U^\nu = -c^2 \tag{11}$$

and therefore the usual mass-shell relation

$$E^2 = (pc)^2 + (mc^2)^2. \tag{12}$$

These are standard consequences of Lorentzian geometry [3, 4, 6]; the vector field u^μ plays no role in this kinematic step.

4 Covariant Action and Field Equations

We take the standard two-derivative Einstein–Æther action [7, 8]

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_u(g_{\mu\nu}, u^\mu, \nabla u)\right] + S_{\rm m}[g_{\mu\nu}, \psi], \tag{13}$$

with

$$\begin{aligned}\mathcal{L}_u = &-c_1(\nabla_\mu u_\nu)(\nabla^\mu u^\nu) - c_2(\nabla_\mu u^\mu)^2 - c_3(\nabla_\mu u_\nu)(\nabla^\nu u^\mu)\\ &- c_4\, u^\mu u^\nu(\nabla_\mu u^\alpha)(\nabla_\nu u_\alpha) + \lambda(u^\mu u_\mu + 1),\end{aligned} \tag{14}$$

where c_i are dimensionless couplings and λ enforces the unit-timelike constraint.

Variation with respect to the metric gives

$$G_{\mu\nu} = 8\pi G\left(T^{(\rm m)}_{\mu\nu} + T^{(u)}_{\mu\nu}\right). \tag{15}$$

Variation with respect to u^μ and λ yields the vector equation of motion and the normalization condition $u^\mu u_\mu = -1$.

By diffeomorphism invariance and the contracted Bianchi identity,

$$\nabla_\mu G^{\mu\nu} \equiv 0, \tag{16}$$

hence

$$\nabla_\mu\left(T^{({\rm m})\mu\nu} + T^{(u)\mu\nu}\right) = 0. \tag{17}$$

Under minimal coupling of matter to the metric alone, the matter Euler–Lagrange equations imply the usual Noether identity

$$\nabla_\mu T^{({\rm m})\mu\nu} = 0. \tag{18}$$

For pressureless dust,

$$T^{(\rm m)}_{\mu\nu} = \rho\, U_\mu U_\nu, \qquad U^\mu U_\mu = -c^2. \tag{19}$$

and eq. (18) implies

$$U^\mu \nabla_\mu U^\nu = 0. \tag{20}$$

It is important to state the scope precisely. In the minimally coupled test-body regime there is no direct non-metric force term in the matter action: the vector field influences motion only through the metric background solving the full coupled field equations. Thus the result is geodesic motion in the metric sector, not the absence of all background-dependent tidal effects in extended or self-gravitating systems. Composition-dependent or Nordtvedt-type deviations can arise beyond the test-body approximation and are naturally described in the PPN framework [9, 6].

5 Weak-Field and PPN Constraints

5.1 Newtonian limit

In the weak-field, slow-motion regime,

$$g_{00} = -\left(1 + \frac{2\Phi}{c^2}\right), \qquad g_{ij} = \delta_{ij}\left(1 - \frac{2\Phi}{c^2}\right), \qquad \left|\frac{\Phi}{c^2}\right| \ll 1, \tag{21}$$

and the (00) component of the field equations reduces to Poisson's equation

$$\nabla^2\Phi = 4\pi G\rho \tag{22}$$

in the GR limit.

5.2 The parameters γ and β

In standard PPN form,

$$g_{00} = -1 + 2U - 2\beta U^2 + \mathcal{O}(\varepsilon^3), \qquad g_{ij} = \delta_{ij}(1 + 2\gamma U) + \mathcal{O}(\varepsilon^2), \tag{23}$$

where $\varepsilon \sim v/c$ and U is the Newtonian potential [5, 6]. For Einstein–Æther theory, Foster and Jacobson show that

$$\gamma = 1, \qquad \beta = 1 \tag{24}$$

for the standard class of asymptotically flat weak-field solutions in which the Æther field is asymptotically aligned with the rest frame of the gravitating source [9]. In the present paper this result is inherited, not re-derived, since the action is the same.

5.3 Preferred-frame parameters

Preferred-frame effects are encoded in α_1 and α_2. We quote the standard Einstein–Æther expressions directly in the original c_i basis following [9]:

$$\alpha_1 = -\frac{8(c_3^2 + c_1c_4)}{2c_1 - c_1^2 + c_3^2}, \tag{25}$$

$$\alpha_2 = \frac{\alpha_1}{2} - \frac{(c_1 + 2c_3 - c_4)(2c_1 + 3c_2 + c_3 + c_4)}{(c_1 + c_2 + c_3)(2 - c_1 - c_4)}. \tag{26}$$

The observational bounds require approximately

$$|\alpha_1| \lesssim 10^{-4}, \qquad |\alpha_2| \lesssim 10^{-7}, \tag{27}$$

which strongly restrict the viable coupling region [6, 8].

5.4 Gravitational-wave speed

The tensor-mode speed is

$$c_{\mathrm{T}}^2 = \frac{1}{1 - c_{13}}. \tag{28}$$

The multimessenger event GW170817 requires

$$|c_{\mathrm{T}} - 1| \lesssim 10^{-15}, \tag{29}$$

hence

$$c_{13} \approx 0 \tag{30}$$

to very high accuracy [10, 11]. In that regime the viable parameter space collapses onto a narrow subspace additionally constrained by Eqs. (25) and (26).

6 Logical Decomposition

The three central conclusions rely on distinct ingredients:

Local SR kinematics. These follow from Lorentzian signature and the existence of local inertial coordinates. Neither the vector dynamics nor the field equations are needed.

Einstein-form field equations. These follow from varying the Einstein–Hilbert sector with respect to the metric. The contracted Bianchi identity constrains the result but is not an independent postulate inserted by hand.

Geodesic free fall in the test-body limit. This follows from minimal matter coupling and the resulting conservation law $\nabla_\mu T^{(\mathrm{m})\mu\nu} = 0$. The statement applies to structureless test bodies and does not by itself exclude higher-order effects for extended, self-gravitating, or compositionally structured systems.

Thus the paper's main value lies in separating three often-conflated claims: Lorentzian kinematics is geometric, Einstein-form dynamics is variational, and test-body WEP behavior is a consequence of minimal coupling.

7 Summary and Outlook

Within a minimal covariant vector–tensor action of Einstein–Æther type, local special-relativistic kinematics, Einstein-form gravitational field equations, and geodesic free fall for minimally coupled structureless test bodies arise from distinct structural assumptions. The paper's contribution is a clean logical synthesis: it identifies which assumptions are doing which conceptual work within the Einstein–Æther framework.

Phenomenologically, the relevant weak-field statements are likewise modest and conditional. The equalities $\gamma = \beta = 1$ are established in [9] for the standard asymptotic-alignment boundary condition, and carry over to the present framework because the action is identical. The main observational pressure then comes from preferred-frame bounds and the gravitational-wave speed constraint $c_{13} \approx 0$.

Further work could sharpen the foundations in three directions: a fully explicit PPN treatment without imposing asymptotic alignment at the outset; a more systematic treatment of self-gravitating bodies and Nordtvedt-type effects; and an analysis of strong-field and cosmological solution sectors.

Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

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